Pilotwings 64 (US) save file (EEPROM)

As of 23 October 2026. From src/app/save.c, code_B2900.c, level_select.c and test_menu.c of the decompilation, checked against a real save file.

File

- 4 kbit EEPROM = 512 bytes (some emulators store 2048; the game uses only the first 512).
- Two saved games of 256 bytes: File 1 at 0, File 2 at 256.

One saved game (256 bytes)

Bytes / bits	Content
bytes 0-1	magic: "PW" = saved game, "pw" = empty (prepared by the game)
from bit 16	results: one 7-bit value per test, 57 tests in the retail game (61 with the game patch bonus-birdman-v1)
behind them	photo album: 6 photos × 172 bits
up to byte 254	unused (zeros)
byte 255	checksum: sum of bytes 0-254, low 8 bits (not set for empty files)

All values are a bit stream; bit n of a byte is 1 << n.

Order of the results: for every vehicle (Hang Glider, Rocket Belt, Gyrocopter, Cannonball, Sky Diving, Jumble Hopper, Birdman), for every class (4 for the license vehicles, 3 for the others), for every test. The number of tests comes from the mission files (UPWT, COMM block: class, vehicle, test), so hacks with other missions get another layout. Retail: license vehicles 1/2/3/3 tests per class; Cannonball 3 levels × 4 targets; Sky Diving and Jumble Hopper 3 levels × 1; Birdman 3 × 4. Birdman stage 4 is not saved.

With the game patch "Bonus games and Birdman" (bonus-birdman-v1): saveFileWrite and saveFileLoad save 3 classes only for Cannonball, Sky Diving and the Jumble Hopper and 4 for all others, so Birdman stage 4 is saved too (4 more results, 28 bits; 61 results). Everything up to Birdman stage 3 stays where it was, but the photo album starts 28 bits later. A save file of the game without the patch therefore shows wrong photos in a patched game (and the other way round); the results are read correctly. The editor's save file tab uses the layout of the open project.

Hacks of Hawk's Hangar (stage 7, Step 41): the results follow the project's missions (up to 128, up to 5 tests per class, classes counted up to 8 per vehicle with the game patch: classes without missions save nothing). Behind the 1032 album bits come **12 extra bits**, two per photo: the third bit of its test and of its class (tests 5 and classes 5–8 need them). With 128 missions the

results take 896 bits, so everything still fits in the 2032 bits. The editor's save tab uses this layout (`SaveGameLayout.FromTasks(…, photoExtras: true)`). **The game does not check the layout:** it reads the bit stream with the layout of its own missions, and the checksum stays right, so a save file written with other missions loads without complaint, but every result behind the first class with another test count is read from the wrong bits (wrong points, unlocks and medals) and the album is shifted (broken photos). When missions are added or removed while a save file is open, the save tab converts it (`EepromFile.ConvertTo` with the `SavePlaceChanges` of the removal: results go along with their missions, so when a test is removed the later tests of the class take their results along to the place in front; the removed mission's result is dropped, new places get 127; photos are kept, a photo of a mission that moved gets its new class and test; the bits between the results and the old or new end are cleared) and marks it as changed, so it can be saved for the new ROM.

Values: 0-100 points (Cannonball: 0-25 per target, 100 per level); 127 = no result yet. Sums count 127 as 0.

Photo record (172 bits): camera position (3 × s16), direction (3 × u8, × 1.40625°), test (2 bits), class (2), object types (4 bits each), animation time per object (u8), vehicle (3), object count (3), object instance (3 bits each). Object types: 1 shuttle, 2 ferry, 3 missile, 4 whale, 5 fountain, 6 flame, 7 whale pod, 8 plane, 9 glider, 10 boat, 11 nothing. Without the fix photo-album-v1 the instance numbers of objects 5 and 6 are lost.

Rules (what is open)

Medal limits (bronze/silver/gold): Beginner 70/80/90, Class A 140/160/180, Class B and Pilot 210/240/270, bonus games 70/80/90 per level. The Birdman's entry (index 7, medal index = vehicle + 1) is 0/0/0 in the game; the game patch bonus-birdman-v1 sets it to 280/320/360 per stage. A license class has a medal only if none of its tests is 127 (in the original game; with own limits from the editor's Classes tab the module and the save tab count only the sums, see Menu_Unlocks.md). Projects can set own limits per vehicle and class (Classes tab); the save tab then uses them.

- **Class A:** all three license vehicles have a medal in Beginner.
- **Class B / Pilot:** per vehicle, a medal in the class before.
- **Cannonball / Sky Diving / Jumble Hopper:** all three license vehicles have silver in Class A / B / Pilot. Level 2 and 3: at least 70 points in the level before.
- **Birdman stage 1:** silver in Beginner with all three vehicles. Stages 2-4: silver in Class A/B/Pilot with all three, and at least 80 points in every level of Cannonball / Sky Diving / Jumble Hopper.

Birdman results open nothing. With the patch, "all perfect" also needs 100 points in every Birdman mission with points (free flights do not count).

The editor's save file tab (Hawk's Hangar) follows these rules. With the patch it shows the points of the 16 Birdman missions (4 stages × 4) below the Birdman row; a stage's fields can be edited when the stage is open.

Own rules (Menu tab, Step 38): a project can give every menu field its own unlock conditions (Menu_Unlocks.md, section 7). The save tab then follows those rules: a field is open when all its conditions are met, and the "Open" button of a closed field raises exactly the fields its conditions name (opening those first if needed) and gives the field's tests without a result 0 points. The Features option "All menu fields open" only changes the game, not the save tab.